AtliQ Hardware

CONSUMER GOODS AD-HOC INSIGHTS

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Request 1:

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

Query:

```
select distinct market
from dim_customer
where customer = "Atliq Exclusive" and region = "APAC"
order by market;
```

Result:

market
Australia
Bangladesh
India
Indonesia
Japan
Newzealand
Philiphines
South Korea

KEY INSIGHTS

- **Pan-APAC Operation:** AtliQ Exclusive sells in 8 APAC markets, including South Asia (India, Bangladesh) to East Asia (Japan, South Korea).
- **Balanced mix:** Four developed markets (Australia, Japan, New Zealand, South Korea) and four developing ones (Bangladesh, India, Indonesia, Philippines) diversify demand.

Request 2:

What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,

unique_products_2020
unique_products_2021
percentage_chg

Query:

```
with unique_2020 as (select count(distinct(product_code)) as unique_products_2020
from fact_sales_monthly
where fiscal_year = 2020),

unique_2021 as (select count(distinct(product_code)) as unique_products_2021
from fact_sales_monthly
where fiscal_year = 2021)

select u.unique_products_2020, un.unique_products_2021,
round((un.unique_products_2021-u.unique_products_2020)*100/u.unique_products_2020, 2) as percentage_chg
from unique_2020 u
join unique_2021 un;
```

Result:

	unique_products_2020	unique_products_2021	percentage_chg
▶	245	334	36.33

KEY INSIGHTS

- **Strong range expansion:** Unique products sold grew from 245 (2020) to 334 (2021) showing a strong 36.33% rise.
- **Net +89 products:** Likely new products returning to sale after the COVID-hit FY20.

Request 3:

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,

segment
product_count

Query:

```
select segment ,count(distinct(product_code)) as product_count
from dim_product
group by segment
order by product_count desc;
```

Result:

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

KEY INSIGHTS

- **Notebook leads:** 129 unique products, about 32% of the 397-product catalogue.
- **Top-3 segments:** Notebook, Accessories and Peripherals hold 329 products, roughly 83% of the range.
- **Bottom-3:** Networking has only 9 products (~2%), the least diversified segment, followed by Storage (27) and Desktop (32).

Request 4:

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,

segment
product_count_2020
product_count_2021
difference

Query:

```
with unique_2020 as (select p.segment, count(distinct(s.product_code)) as unique_products_2020
from fact_sales_monthly s
join dim_product p
using (product_code)
where fiscal_year = 2020
group by p.segment),

unique_2021 as (select p.segment, count(distinct(s.product_code)) as unique_products_2021
from fact_sales_monthly s
join dim_product p
using (product_code)
where fiscal_year = 2021
group by p.segment)

select u.segment, u.unique_products_2020, un.unique_products_2021,
(un.unique_products_2021 - u.unique_products_2020) as difference
from unique_2020 u
join unique_2021 un
using (segment);
```

Request 4: Result

Result:

segment	unique_products_2020	unique_products_2021	difference
Accessories	69	103	34
Desktop	7	22	15
Networking	6	9	3
Notebook	92	108	16
Peripherals	59	75	16
Storage	12	17	5

KEY INSIGHTS

- **Accessories tops the list:** +34 products (69 to 103), 38% of all 89 net new products.
- **Desktop grew fastest:** 7 to 22 products (+214%), the biggest relative jump, although the smallest base.
- **Steady rise:** Notebook (+16, +17%) and Peripherals (+16, +27%) grew moderately; Storage (+5) and Networking (+3) moved the least.

Request 5:

Get the products that have the highest and lowest manufacturing costs.
The final output should contain these fields,

product_code
product
manufacturing_cost

Query:

```
select p.product_code, p.product, m.manufacturing_cost
from dim_product p
join fact_manufacturing_cost m
using (product_code)
where m.manufacturing_cost in ((select max(manufacturing_cost) from fact_manufacturing_cost), (select min(manufacturing_cost) from fact_manufacturing_cost))
order by m.manufacturing_cost desc;
```

Result:

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

KEY INSIGHTS

- **Extreme cost:** AQ HOME Allin1 Gen 2 (240.54) costs ~270x the AQ Master wired x1 Ms (0.89).
- **Dual business model:** A premium high-ticket device versus a low-cost, high-volume accessory.
- **Implication:** Cost control and inventory risk matter most on high-cost products.

Request 6:

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,

customer_code
customer
average_discount_percentage

Query:

```
select c.customer_code, c.customer, ROUND(avg(p.pre_invoice_discount_pct) over(partition by c.customer_code),4) as average_discount_percentage
from dim_customer c
join fact_pre_invoice_deductions p
using (customer_code)
where p.fiscal_year = 2021 and c.market = "India"
order by average_discount_percentage desc
limit 5;
```

Result:

customer_code	customer	average_discount_percentage
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933

KEY INSIGHTS

- **Tight Discount spread:** All top-5 customers get 29.3% to 30.8% pre-invoice discount in FY21.
- **Flipkart leads (30.83%),** followed by Viveks, Ezone and Croma; Amazon is fifth at 29.33%.

Request 7:

Get the complete report of the Gross sales amount for the customer **"Atliq Exclusive"** for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.

The final report contains these columns:

Month

Year

Gross sales Amount

Query:

```
select monthname(s.date) as month, year(s.date) as year, round(sum(s.sold_quantity*g.gross_price),0) as Gross_sales_amount
from fact_sales_monthly s
join fact_gross_price g
using (product_code)
join dim_customer c
using (customer_code)
where c.customer = 'Atliq Exclusive'
group by month, year
order by year
```

Result:

month	year	Gross_sales_amount
September	2019	9092670
October	2019	10378638
November	2019	15231895
December	2019	9755795
January	2020	9584952
February	2020	8083996
March	2020	766976
April	2020	800072
May	2020	1586964
June	2020	3429737
July	2020	5151815
August	2020	5638282
September	2020	19530271
October	2020	21016218
November	2020	32247290
December	2020	20409063
January	2021	19570702
February	2021	15986604
March	2021	19149625

KEY INSIGHTS

- **Festive peak:** November is the strongest month each year with 15.2M (2019) and 32.2M (2020), about 2x YoY growth.
- **COVID impact:** Sales fell ~91% from Feb 2020 (8.08M) to Mar 2020 (0.77M) and stayed under 1.6M until May.

Request 8:

In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,

Quarter

total_sold_quantity

Query:

```
select case
when month(date) in (9,10,11) then "Q1"
when month(date) in (12,1,2) then "Q2"
when month(date) in (3,4,5) then "Q3"
else "Q4"
end as quarters,
sum(sold_quantity) as total_sold_quantity
from fact_sales_monthly
where fiscal_year = 2020
group by quarters
order by total_sold_quantity desc;
```

Result:

quarters	total_sold_quantity
Q1	7005619
Q2	6649642
Q4	5042541
Q3	2075087

KEY INSIGHTS

- **Q1 (Sep-Nov) was the best fiscal quarter:** 7.01M units, about 34% of FY2020 volume.
- **Q1 + Q2 holds 66% of volume:** The festive and year-end season drives most of the demand.
- **Q3 collapse:** Mar-May 2020 sold only 2.08M units (~10%), about 70% below Q1.

Request 9:

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,

channel
gross_sales_mln
percentage

Query:

```sql
with CTE1 as (select c.channel, round(sum(s.sold_quantity*g.gross_price)/1000000, 2) as gross_sales_mln
from fact_sales_monthly s
join dim_customer c
using (customer_code)
join fact_gross_price g
using (product_code)
where s.fiscal_year = 2021
group by c.channel)

select *, round(gross_sales_mln*100/sum(gross_sales_mln) over(), 2) as market_share_pct
from CTE1
order by gross_sales_mln desc;
```

Result:

channel	gross_sales_mln	market_share_pct
Retailer	1924.17	73.22
Direct	406.69	15.48
Distributor	297.18	11.31

KEY INSIGHTS

- **Retailer dominates:** 1,924.17M, or 73.22% of FY2021 gross sales, about 4.7x the Direct channel.
- **Small secondary channels:** Direct (15.48%) and Distributor (11.31%) together contribute only 26.8%.
- **Takeaway:** Relying on one channel is a concentration risk. Growing Direct and Distributor would diversify revenue.

Request 10:

. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,

division
product_code
product, total_sold_quantity, rank_order

Query:

```
with CTE1 as (select p.division, p.product_code, p.product,
sum(s.sold_quantity) as total_sold_quantity
from fact_sales_monthly s
join dim_product p
using (product_code)
where s.fiscal_year = 2021
group by p.division, p.product_code, p.product),

CTE2 as (select *, dense_rank() over(partition by division order by total_sold_quantity desc) as rank_order
from CTE1)

select * from CTE2
where rank_order <= 3;
```

Request 10: Result

Result:

division	product_code	product	total_sold_quantity	rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

KEY INSIGHTS

- **Volume gap among divisions:** N&S top-3 sold 2.07M units and P&A sold 1.27M versus only 52K for PC (about 40x lower).
- **Division Leaders:** AQ Pen Drive 2 IN 1 (701K) in N&S and AQ Gamers Ms (428K) in P&A lead their divisions.